
CIS 6270 · PROJECT 1 DEFENSE · GROUP 2

BDG: Batch-Dispersion Guidance for Property-Band Generation

A feedback-servoed deviation gain for property-gradient guidance, studied on QM9 3D molecules and on DeepFlyBrain DNA enhancers on the probability simplex

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Frame the study

Problem. Steer a frozen generator so that each sample's property lands in a band $y \pm \delta$ around a target. The score is **band coverage**: the fraction of samples whose property falls inside the band.

Question. In gradient guidance, one strength w both moves the batch toward y and shrinks its spread. Does setting the spread separately, by feedback on the batch, raise coverage beyond tuning w ?

Hypothesis. Adjusting dispersion separately from centring raises band coverage at comparable sample quality, with the same rule in both state spaces.

MODALITY 1

QM9 3D molecules [1]

Coordinates in $\mathbb{R}^{N \times 3}$ plus atom types relaxed onto a simplex, $N \leq 29$. Symmetric under translation, rotation and permutation [2].

MODALITY 2

DeepFlyBrain DNA enhancers [4]

500-bp sequences as points of $(\Delta^3)^{500}$: one probability simplex over $\{A, C, G, T\}$ per position. No rotation group; every position must stay on the simplex.

Two state spaces, not two datasets: a Euclidean space acted on by a symmetry group, and a product of probability simplices.

Datasets, splits and leakage

MODALITY 1 · QM9 [1]

133,885 molecules, explicit hydrogens

MoleculeNet release, pinned by SHA-256. We keep the 3,054 molecules that EDM's preprocessing drops [2].

Split	Used for	n
train_a	generator and guide f_A	51,527
train_b	evaluator f_B only	51,527
val	checkpoint selection; molecule sizes	17,748
test	held out	13,083

No leakage. f_B, which scores every sample, never saw a molecule that trained the generator or the guide. Normalisation uses train_a alone.

Preprocessing. QM9: coordinates centred per molecule (Å), padded to 29 atoms with a mask; properties in D, Bohr³ and Ha. DNA: one-hot on the simplex, decoded by per-position argmax.

MODALITY 2 · DEEPFLYBRAIN [4]

Drosophila enhancers, 500 bp

One-hot A, C, G, T; four ambiguous training sequences removed.

Split	Used for	n
train	generator	83,722
val	checkpoint selection	10,505
test	held out	10,434

No learned evaluator. GC and CpG are counted exactly on the decoded sequence, so there is nothing for guidance to exploit and nothing to leak.

Cell-type labels are not given to the generator. A random split says nothing about unseen sequence families.

Representations, conditioning and handling

	QM9 molecules	DeepFlyBrain DNA
Representation	Each atom: 3D coordinates plus a one-hot type over {H, C, N, O, F}. Types are relaxed to continuous values during generation and read back by argmax.	Each position: a probability vector over {A, C, G, T}. A sequence is a point of $(\Delta^3)^{500}$, the simplex view of Dirichlet FM and Fisher Flow [4, 5]; read back by argmax.
Conditioning	None on the property. The only other input is the atom mask, drawn from QM9's size distribution and shared by every arm.	None. Cell-type labels are not given to the generator.
Symmetry and constraints	Translation: noise projected to zero centre of mass. Rotation, reflection, permutation: an E(3)-equivariant EGNN [9]; property nets are invariant. 12/12 equivariance checks pass.	No rotation group. After each step, values are clamped at zero and renormalised per position, so every position stays on the simplex.

Unconditional on purpose. The property enters only at sampling time, through guidance, which is exactly where BDG acts. Both modalities therefore test the same thing.

Flow-matching baseline: the formulation

Flow matching learns a velocity field v_θ that carries noise x_0 to data x_1 along a straight line [6, 7, 8]. Time t runs from 0 (noise) to 1 (data).

Path and target

$$x_t = t x_1 + (1 - t) x_0 \qquad u_t(x_t \mid x_0, x_1) = x_1 - x_0$$

The target is conditioned on the pair (x_0, x_1) , not on any property: v_θ has no condition input.

Noise source

Coordinate noise is projected to zero centre of mass (dimension $3(N - 1)$); type noise is a masked Gaussian.

Loss

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E} \left\| v_\theta(x_t, t) - (x_1 - x_0) \right\|_M^2, \qquad \|z\|_M^2 = \frac{\|M \odot z^c\|^2}{3 \sum M} + \frac{\|M \odot z^f\|^2}{5 \sum M}$$

M is the atom mask. Coordinate (c) and type (f) errors are averaged separately and weighted equally, so padding never dilutes the loss.

Endpoint estimate

$$\hat{x}_1 = x_t + (1 - t) v_\theta(x_t, t)$$

A one-step estimate of the final sample at any t . Guidance differentiates it (2b).

Flow matching: training, selection, sampling

Architecture [9]

E(3)-equivariant graph network (EGNN): width 256, 8 layers, SiLU, 3,753,229 parameters. Time enters through the input embedding; no attention.

Training

train_a only. Adam, learning rate 2×10^{-4} with cosine decay to 5%, batch 256, gradient clip 5, EMA 0.9999. 1,500 epochs (303k steps), 11.4 h on one B200 slice; seed 20260918.

Model selection

Pre-registered: highest validation atom stability at NFE 100, loss as tie-break, every 25 epochs.

Disclosed amendment. The rule picked epoch 1,300; we ship the final EMA weights (epoch 1,500), because 512 validation samples cannot resolve the last 200 epochs. A 3-seed $\times$ 10,000 benchmark agrees: +0.0018 atom, +0.0097 molecule, +0.0110 validity, same sign on every seed.

Sampling [6]

Probability-flow ODE, explicit Euler, 100 uniform steps, so **NFE 100**. Going to 500 steps adds only 0.65 atom and 2.8 molecule points, so 100 steps is the operating point.

NFE (number of function evaluations) counts the network forward passes needed to draw one sample, the standard cost unit for ODE samplers. Flow matching: 100. Our diffusion model: 101 (1d).

Diffusion baseline: the formulation

Variance-preserving (VP) diffusion in continuous time [2, 10, 11]. Noising time u runs from 0 (data) to 1 (noise); the network ε_ϕ predicts the added noise.

Noise schedule	$\beta(u) = \beta_{\min} + u (\beta_{\max} - \beta_{\min}) = 0.1 + 19.9 u$
Forward process	$x_u = a(u) x_1 + \sigma(u) \varepsilon, \quad a(u) = e^{-\frac{1}{2} (0.1 u + 9.95 u^2)}, \quad \sigma^2 = 1 - a^2$
Loss	$\mathcal{L}_{\text{VP}}(\phi) = \mathbb{E} \left\ \varepsilon_\phi(x_u, u) - \varepsilon \right\ _M^2$ The same masked norm as flow matching; ε projected to zero centre of mass; $u \sim \text{U}(10^{-3}, 1)$ keeps σ away from zero.
Sampler	$\frac{dx}{du} = -\frac{1}{2} \beta(u) (x + s_\phi(x, u)), \quad s_\phi = -\varepsilon_\phi / \sigma$ Integrated from $u = 1$ to 0 on a decreasing 100-step grid, then one denoising step to $\hat{x}_1$: NFE 101.
Endpoint estimate	$\hat{x}_1 = (x_u - \sigma(u) \varepsilon_\phi(x_u, u)) / a(u)$ Tweedie's posterior mean: the diffusion counterpart of $x_t + (1 - t)v\theta$, and what guidance differentiates.

Checked: $a^2 + \sigma^2 = 1$ to 6 decimals, $a(0) = 1$, $a(1) = 0.0066$.

Diffusion: training, selection, sampling

Matched by construction. Same representation, split (train_a), EGNN backbone, optimiser, learning-rate schedule, batch, EMA, clip, seed and 1,500-epoch budget as flow matching. Only the corruption process and the output head differ.

Architecture [9]

EGNN 256×8 , 3,753,229 parameters, with a noise (ϵ) head in place of the velocity head.

Model selection

Highest validation stability at NFE 101, loss as tie-break. Flow matching instead ships its final weights (1c): a mismatch we disclose and will align before comparing families.

Sampling [2, 11]

100-step decreasing grid plus the final denoising step: NFE 101 against flow matching's 100. We report the extra evaluation rather than hide it.

Status

Implemented, unit-checked and smoke-tested. The first full run stopped at epoch 80 of 1,500 after a node failure; the full run is pending **[TODO: v3 Job A]**. Until it lands we claim no family winner (2a).

How we will measure the two models

Both base models are ours, and one held-out evaluator scores everything: **f_B**, trained on train_b, with the same band and the same metric code for every model and arm.

Control: band coverage (in-band)

Fraction of decoded samples with $|f_B(x) - y| \leq \delta$. Target y : the property's training median (q50). Band $\delta = 2 \times f_B$'s calibration MAE, fixed before any run.

Fidelity

Molecule stability (every atom's valence satisfied, EDM's rule [2]) and RDKit validity.

Diversity

Uniqueness among valid molecules, and distinct-valid yield (DV): distinct valid molecules per attempt.

Spread

R = standard deviation of f_B over decoded samples $\div$ the unguided value. A diagnostic of what guidance does to the batch, not a diversity measure.

Sample counts. Base models: 3 seeds $\times$ 10,000. Guidance cells: 3 paired seeds (20261001–03), batch 500, $n =$ **[TODO: v3 n]** per cell, a multiple of 500.

What in-band cannot tell us. It inherits f_B 's error, and a guided arm can raise it by exploiting f_B , which is why f_A and f_B train on disjoint halves. It also counts invalid molecules, so it never appears without stability and validity beside it.

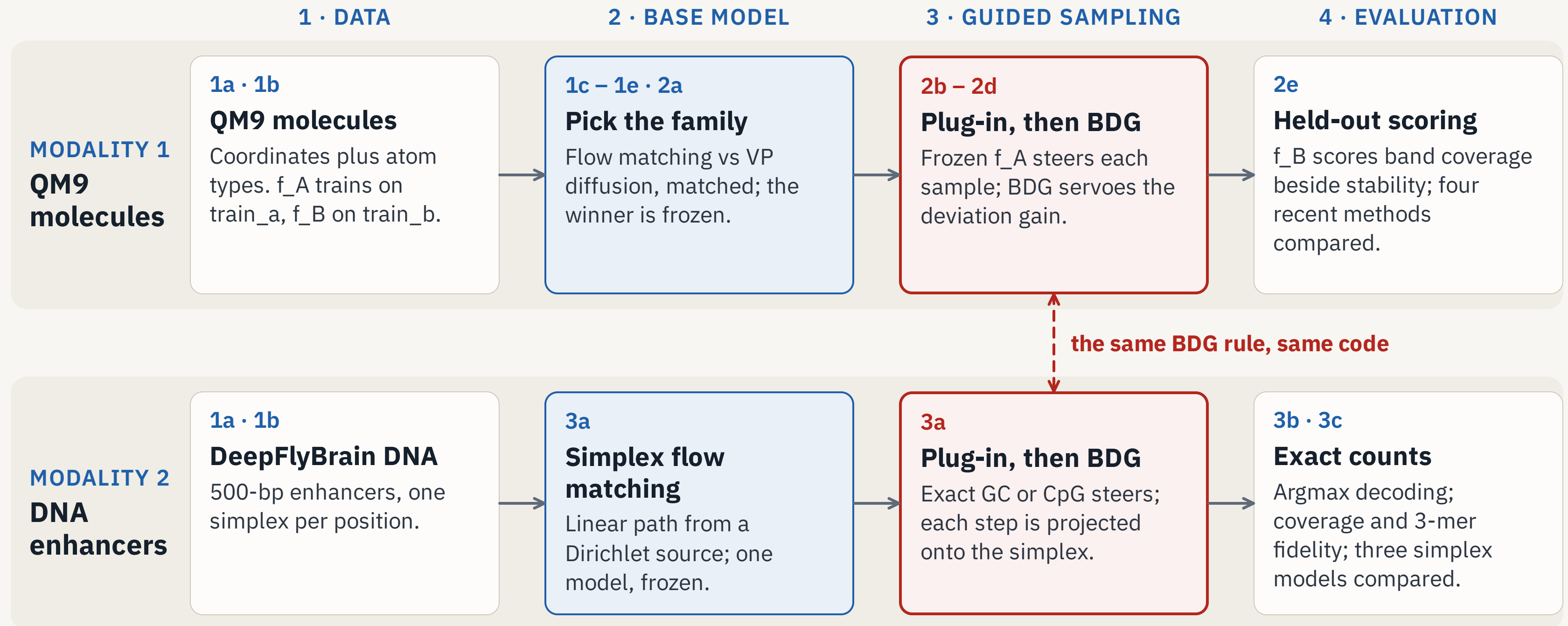
Flow matching vs diffusion: a matched protocol

Both models are ours, trained on the same half of QM9. Everything except the corruption process is held fixed, so 2a compares two families, not two engineering efforts.

Factor	Identical by construction	Necessarily different
Data and split	train_a, 51,527 molecules; same representation	—
Backbone	EGNN 256 × 8, 3.75M parameters each [9]	velocity head vs noise (ϵ) head
Corruption	—	linear interpolant vs variance-preserving: the variable under test
Training recipe	Adam 2×10^{-4} cosine, batch 256, clip 5, EMA 0.9999, seed 20260918, 1,500 epochs	VP run pending [TODO: v3 Job A]
Checkpoint	—	FM: final EMA weights; VP: best validation stability. Disclosed; to be aligned
Sampling	probability-flow ODE, 100 steps	VP adds a denoising step: NFE 101 vs 100
Evaluation	same f_A , f_B , δ , atom masks, seeds and metric code	—

Paired, then tested. Both models start from the same noise, seeds and atom counts, so they are compared sample by sample: seed means, standard deviations and paired per-seed differences over three seeds.

Study overview: one guidance rule, two state spaces



■ trained by us ■ our change ■ data and scoring

Borrowed checkpoints (EquiFM, TFG's predictor pair) are separate comparisons, never pooled with ours.

Choose the base model

Every row is unconditional and unguided — no target, no property gradient. Guidance starts in 2b.

Model	Trained on	Atom stab. ↑	Mol. stab. ↑	Validity ↑	Uniq. ↑	NFE ↓
Flow matching (ours)	train_a (51,527)	0.9366 ±.0012	0.3993 ±.0085	0.7625 ±.0031	0.9939	100
VP diffusion (ours)	train_a (51,527)	[TODO: v3 Job A]	[TODO]	[TODO]	[TODO]	101
EquiFM [24], borrowed	full QM9 (100K)	0.9837	0.8672	0.9453	1.0000	100
Real QM9 — the ceiling	—	0.994	0.956	0.982	—	—

§4.2. Euler probability-flow ODE; our evaluator, EDM's conventions [2]. NFE: network evaluations per sample. Ours: 3 seeds × 10,000. EquiFM: our 256-molecule pilot on the released checkpoint, one seed, trained on twice our data (appendix A1).

Flow matching goes forward, provisionally

The row that settles the family question is still running, and EquiFM cannot stand in for it: an external model, twice the data, a different probability path.

The deciding test is **guided, not unguided**: 2b states it, and the choice stands only if flow matching passes.

Why we sit below the ceiling

1 • Unscaled one-hot features — the published ablation removing exactly that scaling costs 3 atom and 35 molecule points [2].

2 • Training budget — 303k steps against ~1.56M.

Ruled out: the half-data split (~0.3 points [25]) and our evaluator.

Guidance strategy: the plug-in rule

The rule: DPS-style plug-in [12]

$$G_i = \frac{y - F_i}{s^2} J_i^\top \nabla_m f_A(m_i), \quad F_i = f_A(m_i), \quad J_i = \frac{\partial m_i}{\partial x^{(i)}}$$

m_i is the model's one-step estimate of the final sample: $x_t + (1 - t)v_\theta$ for flow matching, the Tweedie mean for diffusion (1c, 1d). y is the property's training median (q50); s is f_A 's training standard deviation.

Frozen at sampling

Generator: ours, frozen.

Guide f_A : ours, trained on train_a; differentiated, never updated.

Evaluator f_B : trained on train_b; never inside the sampler.

How it enters each sampler

$$\text{flow: } \frac{dx}{dt} = v_\theta + \text{clip}_{\|v_\theta\|} \left(w \frac{1-t}{t} G_i \right)$$

$$\text{diffusion: } \frac{dx}{du} = f_\phi + \text{clip}_{\|f_\phi\|} \left(-\frac{1}{2} \beta(u) w G_i \right), \quad f_\phi = -\frac{1}{2} \beta(u) (x + s_\phi)$$

The same G_i , strength $w = 1$ [14], cap and window in both: guidance acts only in the second half of sampling (flow $t \geq 0.5$, diffusion $u \leq 0.5$).

Why the cap

Early in sampling $(1 - t)/t$ is large, and an uncapped correction can overwhelm the model's own velocity: trajectories diverge and coordinates overflow to NaN. In one diagnostic run, **222 molecules broke without the cap and none with it**. The cap is prior practice [13], identical for every arm.

Guided vs unguided, on both base models

Setting. QM9; target $y = q50$; the plug-in rule [12] at $w = 1$ in the second half of sampling; f_A steers, held-out f_B scores ($\delta = 2 \times f_B$ calibration MAE). Each pair shares noise, seeds and atom counts. Registered run: 3 seeds, batch 500 **[TODO: v3]**. Flow-matching cells: pilot, $n = 256$, one seed.

Property	Base model	In-band $\uparrow$	Error / $\delta \downarrow$	Mol. stab. $\uparrow$	Validity $\uparrow$
μ	Flow matching	.098 $\rightarrow$.109	7.56 $\rightarrow$ 5.53	.434 $\rightarrow$.363	.750 $\rightarrow$.707
μ	VP diffusion	[TODO: v3 Job A]	[TODO]	[TODO]	[TODO]
α	Flow matching	.055 $\rightarrow$.051	12.76 $\rightarrow$ 11.08	.434 $\rightarrow$.371	.750 $\rightarrow$.695
α	VP diffusion	[TODO: v3 Job A]	[TODO]	[TODO]	[TODO]
gap	Flow matching	.117 $\rightarrow$.137	5.05 $\rightarrow$ 4.25	.434 $\rightarrow$.336	.750 $\rightarrow$.613
gap	VP diffusion	[TODO: v3 Job A]	[TODO]	[TODO]	[TODO]

Each cell: unguided $\rightarrow$ plug-in. In-band on decoded molecules. Error: mean $|f_B - y|$ in band widths. Uniqueness is 1.000 in every cell.

What the flow model shows

Plug-in pulls the property toward the target: the error falls by **0.8–2.0 band widths** on all three properties (paired $z = -4.2$ to -4.9). At $w = 1$ and $n = 256$ the in-band change is too small to resolve ($|z| \leq 0.84$). The price: 0.06–0.10 molecule stability.

Which property decides the base model

Fixed now, before the diffusion runs land: we lead with the property where plug-in's in-band gain differs most between the two families, with its chemistry cost beside it. Flow matching stays the base model only if it wins there.

Motivating the innovation

The limitation

$$w(y - F_i) = \underbrace{w(y - \bar{F})}_{\text{moves the batch mean}} - \underbrace{w(F_i - \bar{F})}_{\text{shrinks the spread}}$$

In plug-in guidance [12] one strength w scales both terms, so moving the batch toward y also shrinks its spread.

BDG's idea, and the hypothesis

Keep the centring term; set the deviation coefficient by feedback on the batch variance V_b against a requested τ^2 . **Hypothesis:** coverage rises at comparable quality.

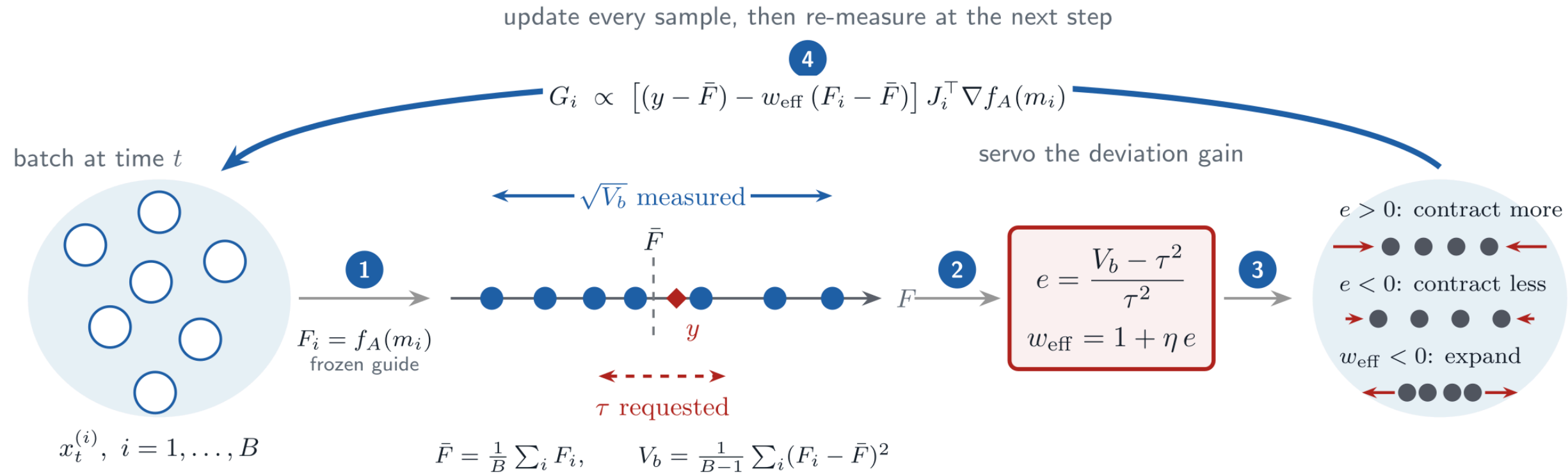
WHERE BDG SITS AMONG THE METHODS COMPARED IN 2E

Method	Statistic it uses	Acts on
DPS-style plug-in [12]	the residual $y - F_i$	one sample
TMPD-inspired [22]	+ the covariance of its own endpoint (a second moment)	one sample
LGD-MC [23]	Monte Carlo draws around the endpoint	one sample
TFG [15]	extra training-free steps on the endpoint	one sample
BDG (ours)	the batch variance of predicted properties, V_b vs τ^2 (a second moment)	the batch

Two different second moments. TMPD weighs one sample's uncertainty about its own endpoint; BDG reads the whole batch's spread in property space. No comparator uses a batch statistic.

Batch-level prior art, not re-run: joint potentials in Particle Guidance [20], moment targets in MGD [18], spread targets in VTD [19], the scale as a control gain in CFG-Ctrl and Feedback Guidance [16, 17], negative-guidance widening [21]. We claim only the composition (bounded search, ~8 queries).

BDG: the mechanism



BDG

$$n_i = \underbrace{(y - \bar{F})}_{\text{centring: fixed at 1}} - \underbrace{w_{\text{eff}}(F_i - \bar{F})}_{\text{deviation: } w_{\text{eff}}, \text{ servoed}}$$

no extra network calls
 $\eta = 0$: plug-in exactly
knobs: $\eta, \tau = \tau_{\text{mult}} s$
both terms scaled by w

One coefficient changes

The centring term keeps coefficient 1; $\eta = 0$ gives plug-in [12] exactly.

Same network calls

Both terms share one gradient; BDG adds two batch reductions.

Two knobs

Gain $\eta \geq 0$ and setpoint $\tau = \tau_{\text{mult}} \cdot s$ (s : f_A 's training s.d.).

BDG: the reduction and its limits

Exact reduction

$$\begin{aligned} n_i &= (y - F_i) - \eta e (F_i - \bar{F}) = (y - \bar{F}) - w_{\text{eff}} (F_i - \bar{F}) \\ &= w_{\text{eff}} (y_{\text{eff}} - F_i), \quad y_{\text{eff}} = \frac{y + \eta e \bar{F}}{w_{\text{eff}}} \quad (w_{\text{eff}} \neq 0) \end{aligned}$$

At any instant BDG is the plug-in direction with a batch-dependent gain w_{eff} and target y_{eff} . It adds no new direction.

When it contracts, and when it expands

$e > 0$ strengthens contraction; $e < 0$ only weakens it. Net expansion needs a negative gain:

$$w_{\text{eff}} < 0 \iff \eta > 1 \text{ and } V_b < \tau^2 (1 - 1/\eta)$$

A bound for any batch

$$V_b \geq 0 \implies e \geq -1 \implies w_{\text{eff}} \geq 1 - \eta$$

So BDG runs two-sided with no clamp.

An idealised picture, not a guarantee

$$\dot{F}_i = (y - \bar{F}) - w_{\text{eff}} (F_i - \bar{F}) \implies \dot{V}_b = -2 w_{\text{eff}} V_b \implies V^* = \tau^2 (1 - 1/\eta) \quad (\eta > 1)$$

Assumes unit response, continuous time and no drift, cap or projection. The stationary variance sits below the setpoint, and nothing guarantees the sampler reaches it.

What we do not claim: independent control of the realised mean and variance. The Jacobian and the cap act on both terms.

BDG in the sampler

Algorithm 1 One BDG-guided Euler step

red: BDG's only change to plug-in

Require: batch x_t of size B ; frozen v_θ and f_A ; target y ; scale s ; gain $\eta \geq 0$; setpoint $\tau > 0$; strength w ; step h ; start $t_{\min}$

- 1: $v \leftarrow v_\theta(x_t, t)$
- 2: **if** $t < t_{\min}$ **return** $\Pi(x_t + h v)$ ▷ guide only for $t \geq t_{\min}$
- 3: $m \leftarrow x_t + (1 - t) v$, with gradients enabled
- 4: $F_i \leftarrow f_A(m_i)$; $\bar{F} \leftarrow B^{-1} \sum_i F_i$
- 5: $V_b \leftarrow (B - 1)^{-1} \sum_i (F_i - \bar{F})^2$
- 6: $e \leftarrow (V_b - \tau^2) / \tau^2$ ▷ one-sided: $e \leftarrow \max(e, 0)$
- 7: $a_i \leftarrow \text{stopgrad} \{ [(y - F_i) - \eta e (F_i - \bar{F})] / s^2 \}$ ▷ coefficients held fixed
- 8: $G \leftarrow \text{VJP}_{x_t}(F, a)$ ▷ one backward pass, as plug-in
- 9: $C_i \leftarrow w (1 - t) / t G_i$
- 10: $C_i \leftarrow C_i \min(1, \|v_i\| / \|C_i\|)$ ▷ velocity-relative cap
- 11: **return** $\Pi\{x_t + h (v + C)\}$ ▷ Π : centring and masks

Checked in code

$\eta = 0$ gives plug-in's field bit for bit; the reduction and the bound hold numerically.

Cost

Plug-in's network calls plus two batch reductions.

The batch is the unit

One batch of 500 is one observation.

Not yet shown: any gain in coverage, quality or transfer. The registered runs decide (2d, 3b) **[TODO: v3 ablation]**.

Testing the innovation: the ablations

Registered grid. $\eta \in \{1, 2, 4, 8\} \times \tau_{\text{mult}} \in \{0.5, 0.75, 1, 1.5\}$, plus $\eta = 0$, each at $w = 1$ and $w = 4$. Three seeds, batch 500, the headline run's n . Every row is paired against plug-in at the same w .

Control	What it isolates	ΔIB vs plug-in $\uparrow$	R	Mol. stab. $\uparrow$
$\eta = 0$	identity: must equal plug-in	[TODO: v3 ablation]	[TODO]	[TODO]
Gain $\times$ setpoint, $w = 1$ and 4	dispersion versus strength	[TODO: v3 ablation]	[TODO]	[TODO]
One-sided, $\max(e, 0)$	the negative-residual branch	[TODO: not yet run]	[TODO]	[TODO]
Open-loop replay of $e(t)$	feedback versus a fixed schedule	[TODO: not yet run]	[TODO]	[TODO]
Signed-strength plug-in	scalar reweighting alone	[TODO: not yet run]	[TODO]	[TODO]

ΔIB : paired change in band coverage from plug-in at the same w . R: decoded property s.d. $\div$ unguided. Rows 1–2 belong to the registered grid; rows 3–5 are additional controls.

Reported with every cell: clip frequency and the realised w_{eff} . A change in coverage is attributed to dispersion only if the cap was not doing the work.

What the ablations can establish

If we observe	We conclude
$\eta = 0$ differs from plug-in	an implementation fault: stop and fix
$\Delta IB > 0$ on all three seeds at comparable stability, and plug-in at $w = 4$ does not match it	BDG raises coverage beyond strength tuning: the full claim
R follows τ_mult over three or more setpoints, but $\Delta IB \approx 0$	BDG sets spread, not coverage: a narrower mechanism claim
open-loop replay matches the closed loop	the effect is a schedule, not feedback
signed-strength plug-in matches BDG	scalar reweighting suffices: we drop the controller claim

Rules fixed in advance

Analyse $w = 1$ and $w = 4$ separately.

Claim a monotone setpoint response only over three or more setpoints.

Check clip frequency and realised gain before crediting feedback.

Treat the batch, not the molecule, as the unit of uncertainty.

Status: every cell is pending **[TODO: v3 ablation]**. A null or a trade-off will be reported as one.

Position against recent work

Method	in-band ↑	Mol. stab. ↑	Diversity ↑	Gen. passes	Guide passes
DPS-style plug-in [12]	[TODO: v3]	[TODO]	[TODO]	10,000	4,000
TMPD-inspired [22]	[TODO: v3]	[TODO]	[TODO]	10,000	4,000
LGD-MC [23]	[TODO: v3]	[TODO]	[TODO]	12,000	20,000
TFG [15]	[TODO: v3]	[TODO]	[TODO]	8,000	20,000
Flow matching, unguided	[TODO: v3]	[TODO]	[TODO]	10,000	0
BDG (ours)	[TODO: v3]	[TODO]	[TODO]	10,000	4,000

§4.5. Every row is reproduced inside our pipeline, on our frozen checkpoint, guide and evaluator, at q50 and w = 1. Network passes per 5,000 molecules.

Why these four

All are inference-time guidance from the last five years, all steer a frozen generator with a frozen predictor, and all run on *our* checkpoint — which isolates **the guidance rule**, the thing our contribution is about.

Efficiency, measured

BDG's two terms share one backward pass: **4,000** guide passes per 5,000 molecules against **20,000** for LGD-MC and TFG. Not the cheapest overall — TFG uses fewer generator passes.

Fair study, and fair discussion

Reproduced, not quoted

Every 2e row is **reproduced by us** on one frozen checkpoint, guide, evaluator, target and sample count. No published score is pooled with ours.

Three port caveats

Plug-in is a DPS-*style* Gaussian-energy field; the original differentiates the residual L2 norm [12].

TMPD is the local-linear scalar limit: a related-work comparison, not a faithful nonlinear port [22].

TFG runs at its published QM9 hyperparameters, but through our deterministic ODE sampler, and our unscaled one-hot features move $\sim 16\times$ less relative to coordinates than the released checkpoint's.

Never read these against TFG's published table [15].

Better, measured

Guide passes: a fifth of LGD-MC's and TFG's (4,000 vs 20,000 per 5,000 molecules), at plug-in's cost.

Worse, measured

Generator passes: TFG needs fewer (8,000 vs 10,000).

Pending

Coverage and chemistry against every method, per property, from the registered run **[TODO: v3]**.

The transferred method

Unchanged: the rule. $n_i, \bar{F}, V_b, e, \tau = \tau_{\text{mult}} \cdot s, \eta$ and $w_{\text{eff}} = 1 + \eta e$ carry over as written: one scalar per sample times the plug-in step, so no covariance matrix or inverse is needed (the simplex covariance $\text{diag}(p) - pp^T$ is singular).

What changes	Modality 1 · QM9	Modality 2 · DeepFlyBrain
State	$\mathbb{R}^{N \times 3}$ plus a type simplex	$(\Delta^3)^{500}$
Generator	EGNN flow matching, 3.75M	dilated 1-D CNN, 128×10 layers, $\approx 1.02\text{M}$; linear path from a Dirichlet(1, 1, 1, 1) source [4]
Guide	learned f_A	soft GC fraction or CpG frequency (analytic)
Evaluator	learned f_B	exact GC or CpG count after argmax
Feasibility	centring and atom masks	clamp at zero, renormalise each position

Training
83,722 sequences; 1,500 epochs (492k steps), batch 256, learning rate 2×10^{-3} with cosine decay; best validation loss 0.0634.

Sampling
Euler with a projection after every step, then argmax. Window, batch, n and seeds declared before the run **[TODO: v3 Job B]**.

$$F_{\text{GC}}(p) = \frac{1}{L} \sum_{j=1}^L (p_{j,\text{C}} + p_{j,\text{G}}), \quad F_{\text{CpG}}(p) = \frac{1}{L-1} \sum_{j=1}^{L-1} p_{j,\text{C}} p_{j+1,\text{G}}$$

GC is affine in p and CpG is quadratic, so the pair tests whether BDG depends on the property's curvature.
Known gap: the guide reads an endpoint that can leave the simplex; the evaluator counts decoded bases.

Transfer results

DeepFlyBrain, length 500; GC and CpG reported separately; one generator, window and band for every guided row.

Method	In-band ↑	3-mer JSD ↓	Diversity ↑	Time ↓
Simplex FM, unguided	[TODO: v3 Job B]	[TODO]	[TODO]	[TODO]
Simplex FM + plug-in	[TODO: v3 Job B]	[TODO]	[TODO]	[TODO]
Simplex FM + BDG (ours)	[TODO: v3 Job B]	[TODO]	[TODO]	[TODO]
Dirichlet FM [4]	[TODO: v3 Job B]	[TODO]	[TODO]	[TODO]
Fisher Flow [5]	[TODO: v3 Job B]	[TODO]	[TODO]	[TODO]
MOG-DFM [3]	[TODO: v3 Job B]	[TODO]	[TODO]	[TODO]

In-band: exact count inside the declared band. JSD: 3-mer mismatch with held-out sequences. Diversity: mean normalised pairwise Hamming distance.

Fair study

All rows share length, target, band, conditioning, decoding and reference data. Rows 4–6 are the published models, not our guidance ports; we quote none of their class-conditioned scores, nor the enhancer classifier Fisher Flow questions [5].

Fair discussion

Exact counts carry no evaluator error, but GC is coarse, so coverage is read beside 3-mer fidelity and diversity. The band is declared, not measured, so DNA and QM9 coverage are never compared.

Does the innovation transfer?

Criterion, fixed in advance. The innovation transfers if BDG's change from plug-in has the same sign in both modalities, in coverage and in how spread follows the setpoint, at a comparable fidelity cost. Each modality is judged against its own plug-in.

Modality	Property	Δ IB vs plug-in	Spread vs setpoint	Fidelity cost
QM9	μ , α , gap	[TODO: v3]	[TODO]	Δ mol. stab. [TODO]
DNA	GC	[TODO: v3 Job B]	[TODO]	Δ 3-mer JSD [TODO]
DNA	CpG	[TODO: v3 Job B]	[TODO]	Δ 3-mer JSD [TODO]

What would count against transfer

A gain in one modality only, or spread that follows the setpoint in one modality only.

Why both GC and CpG

GC is affine in the simplex coordinates and CpG is quadratic. If BDG helps on one only, curvature, not the rule, explains it.

Status

No transfer claim until both runs land. Raw coverage is never compared across modalities: their bands follow different rules.

Limitations and failure modes

1 · The evaluator bounds the claim

In-band inherits f_B 's error, and guidance is an optimiser that can find f_B 's blind spots. Disjoint training halves reduce this risk; they do not remove it.

2 · What is guided is not what is scored

The guide reads a relaxed endpoint (continuous atom types, or DNA off the simplex) while scoring uses decoded samples. The gap can hide an effect or fake one.

3 · The cap and the projection edit the correction

A velocity cap or a simplex projection can truncate BDG's term, and a cap-limited change is not a controller effect. Hence clip frequency and realised w_{eff} are reported with every cell.

4 · The batch is the estimator

$\bar{F}$ and V_b couple the samples, so results depend on the batch size (500), and uncertainty must be computed over batches, not molecules.

Also open: the base-model choice (our diffusion model is not yet trained); equal nominal w is not equal applied force across arms; the four external methods are adaptations run in our pipeline.

Conclusion and future directions

Does setting dispersion separately from centring raise band coverage at comparable quality?

Not yet shown.

The registered runs decide. **A null or a trade-off will be reported as one.**

What we can already say

BDG changes **one coefficient**: the deviation gain, set by feedback on the batch variance.

Algebraically it is **plug-in with a batch-dependent gain and target**.

It expands the batch **only when $\eta > 1$ and $V_b < \tau^2(1 - 1/\eta)$** .

It costs **plug-in's network calls**, in both state spaces.

What would withdraw the claim

Signed-strength plug-in matching BDG at equal chemistry (**reweighting suffices**), or open-loop replay matching the closed loop (**a schedule, not feedback**).

Next experiments

- 1 • Train our diffusion model; run the guided FM-vs-VP test (Job A).
- 2 • The registered 3-seed ablation grid.
- 3 • DeepFlyBrain on GC and CpG against three simplex models (Job B).

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APPENDIX · NOT PART OF THE 15-MINUTE TALK

Q&A backup

Four topics, one or two slides each.

A1

EquiFM, the borrowed generator

Architecture, path, and how our arms run on it

A2

Property predictors

Ours vs TFG's: architecture, training, accuracy

A3

The data split

Who trains on which half, and why halves

A4

The guidance arms

Each arm's correction, what it changes, what it costs

EquiFM: the borrowed generator

Released checkpoint	
Paper	Song et al., NeurIPS 2023 [24]
Backbone	EDM-style equivariant GNN dynamics [2, 9], 256×9
Parameters	5,339,920 (EMA)
State per atom	3 coordinates (Å) + 5 one-hot ÷ 4 + charge ÷ 10; explicit H
Training data	the full 100K QM9 partition, unconditional

Hybrid probability path

Native time τ : 1 = noise, 0 = data.

Coordinates: almost linear — but the noise is rotated (Kabsch) and permuted (Hungarian) onto the molecule first, an equivariant optimal-transport coupling, plus an angle penalty.

Types and charge: variance-preserving, β from 0.1 to 20.

Sampling: Euler, $\tau = 1 \rightarrow 0$.

$$x_\tau = (1 - \tau) x_0 + s_x(\tau) z, \quad s_x = \epsilon + (1 - \epsilon)\tau$$

$$h_\tau = a h_0 + \sqrt{1 - a^2} z', \quad a = e^{-T(\tau)/2}$$

Our unguided pilot

256 molecules, 100 Euler steps: atom **98.37%**, molecule **86.72%**, validity **94.53%**, uniqueness 100%. A usability check, not a benchmark.

Why its chemistry beats ours

Four candidates, none isolated by us: $\sim 2\times$ the training data, 5.34M vs 3.75M parameters, one-hot scaled by $\frac{1}{4}$ (our leading suspect in 2a), and the aligned coordinate path.

Running our guidance arms on EquiFM

Why it is in the study: a second, external flow-matching generator asks whether the comparison survives on a model we did not train. Backend **equifm**, all seven arms [24].

Endpoint estimate m	$m_x = (1 - \epsilon) x_\tau - s_x q_x, \quad m_h = a h_\tau + q_h/a$	Exact — algebra from its own regression targets.
Covariance Σ TMPD and LGD-MC only	$\Sigma = KJ, \quad k_x = \frac{s_x^2}{1 - \tau}, \quad k_h = \frac{1 - a^2}{a}$	Exact for types, approximate for coordinates , whose aligned noise is not an independent Gaussian channel.
Guided velocity	$\Delta v_x = -\frac{s_x}{1 - \tau} G_x, \quad \Delta v_h \approx -\frac{\beta(\tau)}{2} G_h$	The model's own velocity map at the guided mean $m + K \cdot G$.

Property pair here

TFG's released guide (tf_predict at time 0) and oracle (evaluate) [15]. Its band is narrower than ours — δ 0.157 vs 0.175 (μ), 0.165 vs 0.506 (α), 0.0037 vs 0.0075 (gap) — so **in-band never crosses backends**.

Disclosed limits

Trained on the full 100K, so it saw both halves. Only TMPD and LGD-MC touch the Σ surrogate; plug-in, TFG and BDG do not. The charge channel is held fixed during guidance. 1.85× our memory per molecule.

Property predictors: our pair and TFG's

Ours score the **fm** and **EDMsecond** backends; TFG's score **EquiFM**. f_A's **gradient** steers sampling; f_B only **scores**.

	Ours — f_A and f_B	TFG guide (tf_predict)	TFG oracle (evaluate)
Network	invariant EGNN on squared distances, mean-pool, linear head [9]	EDM dynamics EGNN with an energy head [2, 15]	EDM property-classifier EGNN [2]
Width × layers	128 × 4	192 × 7	128 × 7
Edge attention	no — equal-weight sum	yes	yes
Parameters (μ)	429,825 each	1,671,566	743,944
Inputs	coordinates + unscaled one-hot	+ one-hot ÷ 8; time fixed at 0	+ nuclear-charge powers
Training data	f_A: train_a · f_B: train_b (51,527 each)	QM9 second half (~50K)	QM9 first half (~50K)
Recipe	clean molecules, MSE on a standardised target; Adam 3e-4 cosine, batch 128, 120 epochs (~23 min); best validation MAE	clean molecules; EMA at epoch 2,000	best-validation checkpoint

Theirs are better; ours are disjoint

MAE	μ (D)	α (Bohr ³)	gap (Ha)
our f_A	0.0872	0.2325	0.00402
our f_B	0.0877	0.2531	0.00374
TFG guide	0.0417	0.0925	0.00185
TFG oracle	0.0784	0.0824	0.00187

Same 3,000 calibration molecules. Ours are scored only on the half each did not train on; TFG's on the whole pool, which overlaps their training halves and flatters them.

The honest gap on μ . On molecules neither TFG net trained on: guide 0.0659 D, oracle 0.0728 D, our f_B 0.0840 D. **~20% better, not 2×.**

Why theirs are more accurate
Capacity: 2–4× the parameters, 7 layers not 4.
Attention: a learned weight per neighbour; ours sums neighbours equally.
Training: the guide is an EMA at epoch 2,000 [15]; ours stop at 120.
Inputs: the oracle also reads nuclear-charge features.

Why we keep ours on our FM
Ours is disjoint **by construction**; TFG's only **by inference** — its saved arguments are overwritten by its own training script, and the oracle's training IDs are unknown. The price is a wider band (ours / TFG δ : 1.12× μ , 3.07× α , 2.00× gap), which is exactly why in-band never crosses backends.

Disjointness test (μ , 3,000 molecules): guide wins by >2× on 30.9%, oracle on 40.9%, neither on 28.2% — against 37.4 / 37.4 / 25.3% predicted for disjoint halves. Replicated on α and gap.

The data split, and why it has this shape

Split	Trains or supplies	n
train_a	generator (FM and VP) and guide f_A	51,527
train_b	evaluator f_B only	51,527
val	checkpoint selection; molecule sizes	17,748
test	never trained or selected on	13,083

δ calibration: 3,000 molecules from train_a + train_b; each of our nets is scored only on the half it did not train on.

QM9 [1], 133,885 molecules, pinned by SHA-256; split fixed by seed 20260917.

1 · f_A and f_B never share data

Guidance is an optimiser aimed at f_A; pushed hard it finds inputs that fool it. The checker needs different blind spots, so the guide-evaluator gap stays a usable reward-hacking signal.

2 · The generator may share with f_A

It judges nothing. This keeps the claim to one sentence: **f_B never saw a molecule the generator or the guide trained on.**

3 · Halves, as the literature does

EDM's conditional protocol and TFG's EDMsecond train on one ~50K half; TFG shipped a full-data EDMfull and *chose* EDMsecond [2, 15]. Using both halves would set twice the data beside every number we compare to. 70/30 was rejected for the same reason.

The cost we accept: half the data. A published half-data EDM loses only ~0.3 atom-stability points [2, 25] — so the split does not explain our gap in 2a.

The guidance arms, side by side

Shared frame. $m = x_t + (1 - t)v\theta \rightarrow$ frozen $f_A \rightarrow$ pull back through $J = \partial m / \partial x_t \rightarrow$ clip $\rightarrow$ add to the velocity ($t \geq 0.5, w = 1$).

Arm	Correction	What it changes	Passes per 5,000 (2e)
DPS-style plug-in [12]	$G = \frac{y - f_A(m)}{s^2} J^\top \nabla f_A(m)$	the property at the endpoint; the baseline	10,000 / 4,000
TMPD-inspired [22]	$G = \frac{y - f_A(m)}{s^2 + g^\top \Sigma g} J^\top g, \quad \Sigma = k J, \quad g = \nabla f_A(m)$	shrinks the step where the endpoint is uncertain	10,000 / 4,000
LGD-MC [23]	$G = J^\top \sum_k \omega_k \frac{y - f_A(m + rz_k)}{s^2} \nabla f_A(m + rz_k), \quad r^2 = \frac{1}{d} \text{tr } \Sigma$	averages the likelihood over K draws around m	12,000 / 20,000
TFG [15]	$L(x) = \log \frac{1}{K} \sum_k \exp \left[- \left(\frac{f_A(x + \epsilon_k) - y}{\text{MAD}} \right)^2 \right]$ $\Delta_t = \rho \nabla_{x_t} L(m), \quad \Delta_0 : m \leftarrow m + \mu \nabla_m L(m) \quad (4 \text{ steps, no } J)$	adds steps on the predicted molecule itself	8,000 / 20,000
BDG (ours)	$\text{num}_i = (y - \bar{F}) - w_{\text{eff}} (F_i - \bar{F}), \quad w_{\text{eff}} = 1 + \eta e$	plug-in with a servoed deviation gain	10,000 / 4,000